Multidimensional features with linear morphology

The case for an independent morphological level of representation

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Seri (isolate)

- Seri is spoken on the coast of Sonora (Mexico) in two villages: El Desemboque/Haxöl Iihom and Punta Chueca/Socaaix

- It is spoken by approximately 900 speakers (Ethnologue 2007)

THE PROBLEM

The problem

- Plural suffixes in nouns and verbs, with subtler meanings

PL MARKERS IN NOUNS		PL MARKERS IN VERBS	
paar-oj	'priests'	**itacsipx-oj**	's/he glues multiple times'
		tmasl-oj	'it is black in various places'
		itzacoz-oj	'they peck'
		tacöt-oj	'they cover multiple times'
haahol-cam	'roads'	**toeequel-cam**	'it teeters on the edge multiple times'
		itapooil-cam	'they twist'
		itapocal-cam	'they remove its shell multiple times'
xpaahjö-lca	'hematites'	**itamoosa-lca**	'it is creased in various places'
		ittiipxözi-lca	'they tighten'
		itahoaatjö-lca	'they sweeten multiple times'
nop-xam	'bobcats'	**itsap-xam**	'they swat at'
		ititooc-xam	'they sow multiple times'

Examples from Moser & Marlett (2010)

The problem

- Polyfunctionality: each exponent can express a range of meanings across verbs

itsap-xam	'they swat at'	**ititooc**	'they sow'	
itsap-tolca	'they swat at'	**ititooc-xam**	'they sow'	(multiple times)
itapooil-cam	'they twist'	**itapocl-coj**	'they remove its shell'	
itapooiz-olca	'they twist'	**itapocal-cam**	'they remove its shell'	(multiple times)
itzacoz-oj	'they peck'	**tacöt-am**	'they cover'	
itzacoz-am	'they peck'	**tacöt-oj**	'they cover'	(multiple times)

Examples from Moser & Marlett (2010)

The problem

- Syncretism: each exponent can express a range of meanings within a paradigm

cöitfiz-xam	'they make knots'
cöitf<coo>z-xam	'they make knots (multiple times)'
itazaail-cam	'they anchor'
itazaiil-cam	'they anchor (multiple times)'
cöitaafip-oj	's/he unfastens (multiple times)'
cöita<too>fip-oj	'they unfasten'

Examples from Moser & Marlett (2010)

FEATURE SYSTEM

Seri verbs

- Number distinctions in verbs; see Marlett (1988, 2016) and Moser & Marlett (2010)
  - Subject number
    - Singular
    - Plural
  - Verbal number
    - Neutral
    - Multiple; see Cabredo Hofherr, Pasquereau, O'Meara (2018)
    - Distributional (Moser 1961's "sequential", not explicitly factored into subsequent work); see Pasquereau (2019) for arguments that it does not require temporal distribution

Seri verbs: subject number

- Singular subject

Moxima *sahmees* *hizcoi* ***h-yoohit.***
yesterday orange DEM.PL 1SG-RLS.YO.eat
'Yesterday I ate these oranges.'

Moxima *sahmees* *hizcoi* ***h-yoohitim.***
yesterday orange DEM.PL 1SG-RLS.YO.eat.MULT
'Yesterday I ate these oranges (over time).'

* *Moxima* *sahmees* *hizcoi* ***h-yoiitoj.***
yesterday orange DEM.PL 1SG-RLS.YO.eat.PL
Int. 'Yesterday I ate these oranges.'

* *Moxima* *sahmees* *hizcoi* ***h-yoiitolca.***
yesterday orange DEM.PL 1SG-RLS.YO.eat.PL.MULT
Int. 'Yesterday I ate these oranges (over time).'

Seri verbs: subject number

- Plural subject

* *Moxima* *sahmees* *hizcoi* ***ha-yoohit.***
yesterday orange DEM.PL 1PL-RLS.YO.eat
Int. 'Yesterday we ate these oranges.'

* *Moxima* *sahmees* *hizcoi* ***ha-yoohitim.***
yesterday orange DEM.PL 1PL-RLS.YO.eat.MULT
Int. 'Yesterday we ate these oranges (over time).'

Moxima *sahmees* *hizcoi* ***ha-yoiitoj.***
yesterday orange DEM.PL 1PL-RLS.YO.eat.PL
'Yesterday we ate these oranges.'

Moxima *sahmees* *hizcoi* ***ha-yoiitolca.***
yesterday orange DEM.PL 1PL-RLS.YO.eat.PL.MULT
'Yesterday we ate these oranges (over time).'

Seri verbs: event number

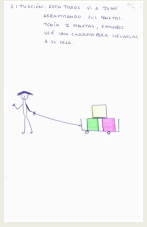
- Neutral form is underspecified

Juan *quih* *xiica an iqueaacalca* *coi* *hant* ***iyootox***
Juan DEF suitcases DEF.PL down 3>3.RLYO.extend
'Juan dragged the suitcases.'

Context 1: true
Distribution over times: ✓
Distribution over spaces: ?

Context 2: true
Distribution over times: ✕
Distribution over spaces: ✓

Context 3: true
Distribution over times: ✕
Distribution over spaces: ✕



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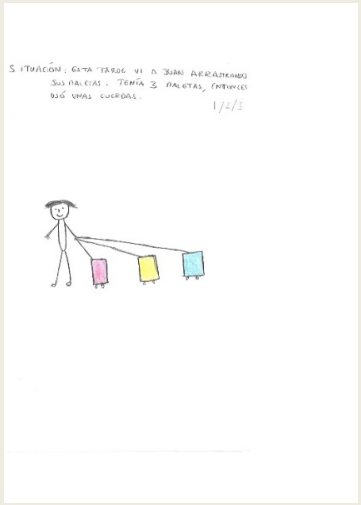
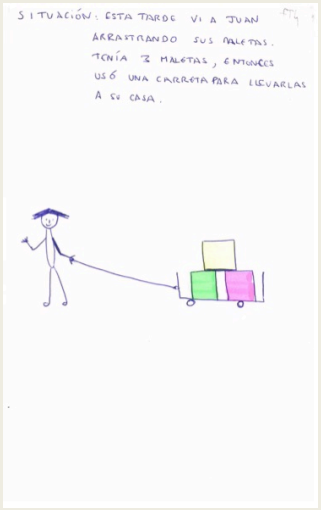


Context 2: true
Distribution over times: ✕
Distribution over spaces: ✓

Seri verbs: event number

- Neutral form is underspecified

Juan *quih* *xiica an iqueaacalca*
Juan DEF suitcases
coi ***hant*** ***iyootox***
DEF.PL down 3>3.RLYO.extend
'Juan dragged the suitcases.'



Context 3: true
Distribution over times: ✕
Distribution over spaces: ✕

Seri verbs: multiple

- Multiple form requires multiple events which are, at least, distributed over times

Juan *quih* *xiica an iqueaacalca* *coi* ***hant*** ***iyootoxim***
Juan DEF suitcases DEF.PL down 3>3.RLYO.extend.MULT
'Juan dragged the suitcases.'

Context 1: true
Distribution over times: ✓
Distribution over spaces: ?

Context 2: false
Distribution over times: ✕
Distribution over spaces: ✓

Context 3: false
Distribution over times: ✕
Distribution over spaces: ✕

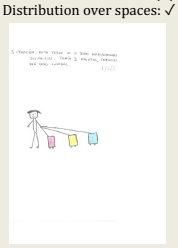
Seri verbs: distributional

- Distribution form requires multiple events which are, at least, distributed over spaces

Juan *quih* *xiica an iqueaacalca* *coi* ***hant*** ***iyootyax***
Juan DEF suitcases DEF.PL down 3>3.RLYO.extend.DIST
'Juan dragged the suitcases.'

Context 1: true
Distribution over times: ✓
Distribution over spaces: ?

Context 2: true
Distribution over times: ✕
Distribution over spaces: ✓



Context 3: false
Distribution over times: ✕
Distribution over spaces: ✕

Seri verbs

- Two cross-classifying features
  - Subject number [singular, plural]
  - Verbal number [neutral, multiple, distributional]
    (previously only 2 values had been recognized)

subject number	'wrap'	event number: neutral	multiple	distributional
	singular	**iyacapnij**	**iyacapanl**	**iyacapnalca**
	plural	**iyacapnalcoj**	**iyacapzil**	**iyacapzilca**

- In principle, 6 cells but in the overwhelming majority of cases, if there is a separate distributional form, it is in the singular subject paradigm

subject number	'drag'	event number: neutral	multiple	distributional
	singular	**hant iyootox**	**hant iyootoxim**	**hant iyootyax**
	plural	**hant iyootyajc**	**hant iyootyaxlca**	

FEATURE SYSTEM

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	-taxanl		-taxnal-ca	-taxnal-coj		'scold'
-atox	-atoxim	-atyax	-atyaj-c	-atyax-lca		'stretch'
-tahoiij		-tahoiil-c	-tahoiil-coj			'make tubular'
-tzam	-tzam-tim	-tzamlqu-im	-tzamlc-am			'corner, attack'
-tanamj	-tanaml-c		-anaml-coj	-anaml-cam		'hurry'
-tazaain-im	-tazaiin-im		-azaail-cam	-azaiil-cam		'anchor'
-tpoc	-tpoct-im		-tpocl-im	-tpocal-am		'fall'
-tpazj-c	-tpaxlax		-tpazlax	-tpazlax-lca		'be scattered'
-tineezil-ca	-tineezil-im		-tineezil-coj	-tineezil-am		'be raspy'
-tapox	-tapox-im	-taptax	-taptaj-c	-tapo-tam	-taptax-lca	'pull out'

-tim

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	-taxanl		-taxnal-ca	-taxnal-coj		'scold'
-atox	-atoxim	-atyax	-atyaj-c	-atyax-lca		'stretch'
-tahoiij		-tahoiil-c	-tahoiil-coj			'make tubular'
-tzam	-tzam-tim	-tzamlqu-im	-tzamlc-am			'corner, attack'
-tanamj	-tanaml-c		-anaml-coj	-anaml-cam		'hurry'
-tazaain-im	-tazaiin-im		-azaail-cam	-azaiil-cam		'anchor'
-tpoc	-tpoct-im		-tpocl-im	-tpocal-am		'fall'
-tpazj-c	-tpaxlax		-tpazlax	-tpazlax-lca		'be scattered'
-tineezil-ca	-tineezil-im		-tineezil-coj	-tineezil-am		'be raspy'
-tapox	-tapox-im	-taptax	-taptaj-c	-tapo-tam	-taptax-lca	'pull out'

-c

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	-taxanl		-taxnal-ca	-taxnal-coj		'scold'
-atox	-atoxim	-atyax	-atyaj-c	-atyax-lca		'stretch'
-tahoiij		-tahoiil-c	-tahoiil-coj			'make tubular'
-tzam	-tzam-tim	-tzamlqu-im	-tzamlc-am			'corner, attack'
-tanamj	-tanaml-c		-anaml-coj	-anaml-cam		'hurry'
-tazaain-im	-tazaiin-im		-azaail-cam	-azaiil-cam		'anchor'
-tpoc	-tpoct-im		-tpocl-im	-tpocal-am		'fall'
-tpazj-c	-tpaxlax		-tpazlax	-tpazlax-lca		'be scattered'
-tineezil-ca	-tineezil-im		-tineezil-coj	-tineezil-am		'be raspy'
-tapox	-tapox-im	-taptax	-taptaj-c	-tapo-tam	-taptax-lca	'pull out'

-ca

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	-taxanl		-taxnal-ca	-taxnal-coj		'scold'
-atox	-atoxim	-atyax	-atyaj-c	-atyax-lca		'stretch'
-tahoiij		-tahoiil-c	-tahoiil-coj			'make tubular'
-tzam	-tzam-tim	-tzamlqu-im	-tzamlc-am			'corner, attack'
-tanamj	-tanaml-c		-anaml-coj	-anaml-cam		'hurry'
-tazaain-im	-tazaiin-im		-azaail-cam	-azaiil-cam		'anchor'
-tpoc	-tpoct-im		-tpocl-im	-tpocal-am		'fall'
-tpazj-c	-tpaxlax		-tpazlax	-tpazlax-lca		'be scattered'
-tineezil-ca	-tineezil-im		-tineezil-coj	-tineezil-am		'be raspy'
-tapox	-tapox-im	-taptax	-taptaj-c	-tapo-tam	-taptax-lca	'pull out'

-lca

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	-taxanl		-taxnal-ca	-taxnal-coj		'scold'
-atox	-atoxim	-atyax	-atyaj-c	-atyax-lca		'stretch'
-tahoiij		-tahoiil-c	-tahoiil-coj			'make tubular'
-tzam	-tzam-tim	-tzamlqu-im	-tzamlc-am			'corner, attack'
-tanamj	-tanaml-c		-anaml-coj	-anaml-cam		'hurry'
-tazaain-im	-tazaiin-im		-azaail-cam	-azaiil-cam		'anchor'
-tpoc	-tpoct-im		-tpocl-im	-tpocal-am		'fall'
-tpazj-c	-tpaxlax		-tpazlax	-tpazlax-lca		'be scattered'
-tineezil-ca	-tineezil-im		-tineezil-coj	-tineezil-am		'be raspy'
-tapox	-tapox-im	-taptax	-taptaj-c	-tapo-tam	-taptax-lca	'pull out'

-coj

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	-taxanl		-taxnal-ca	-taxnal-coj		'scold'
-atox	-atoxim	-atyax	-atyaj-c	-atyax-lca		'stretch'
-tahoiij		-tahoiil-c	-tahoiil-coj			'make tubular'
-tzam	-tzam-tim	-tzamlqu-im	-tzamlc-am			'corner, attack'
-tanamj	-tanaml-c		-anaml-coj	-anaml-cam		'hurry'
-tazaain-im	-tazaiin-im		-azaail-cam	-azaiil-cam		'anchor'
-tpoc	-tpoct-im		-tpocl-im	-tpocal-am		'fall'
-tpazj-c	-tpaxlax		-tpazlax	-tpazlax-lca		'be scattered'
-tineezil-ca	-tineezil-im		-tineezil-coj	-tineezil-am		'be raspy'
-tapox	-tapox-im	-taptax	-taptaj-c	-tapo-tam	-taptax-lca	'pull out'

-tam

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	**-taxanl**		**-taxnal-ca**	**-taxnal-coj**		'scold'
-atox	**-atoxim**	**-atyax**	**-atyaj-c**	**-atyax-lca**		'stretch'
-tahoiij		**-tahoiil-c**	**-tahoiil-coj**			'make tubular'
-tzam	**-tzam-tim**	**-tzamlqu-im**	**-tzamlc-am**			'corner, attack'
-tanamj	**-tanaml-c**		**-anaml-coj**	**-anaml-cam**		'hurry'
-tazaain-im	**-tazaiin-im**		**-azaail-cam**	**-azaiil-cam**		'anchor'
-tpoc	**-tpoct-im**		**-tpocl-im**	**-tpocal-am**		'fall'
-tpazj-c	**-tpaxlax**		**-tpazlax**	**-tpazlax-lca**		'be scattered'
-tineezil-ca	**-tineezil-im**		**-tineezil-coj**	**-tineezil-am**		'be raspy'
-tapox	**-tapox-im**	**-taptax**	**-taptaj-c**	**-tapo-tam**	**-taptax-lca**	'pull out'

-cam

One inventory of exponents for all feature values

- For every exponent$_x$, it is not possible to associate it$_x$ with one bundle of feature values

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-taxnij	**-taxanl**		**-taxnal-ca**	**-taxnal-coj**		'scold'
-atox	**-atoxim**	**-atyax**	**-atyaj-c**	**-atyax-lca**		'stretch'
-tahoiij		**-tahoiil-c**	**-tahoiil-coj**			'make tubular'
-tzam	**-tzam-tim**	**-tzamlqu-im**	**-tzamlc-am**			'corner, attack'
-tanamj	**-tanaml-c**		**-anaml-coj**	**-anaml-cam**		'hurry'
-tazaain-im	**-tazaiin-im**		**-azaail-cam**	**-azaiil-cam**		'anchor'
-tpoc	**-tpoct-im**		**-tpocl-im**	**-tpocal-am**		'fall'
-tpazj-c	**-tpaxlax**		**-tpazlax**	**-tpazlax-lca**		'be scattered'
-tineezil-ca	**-tineezil-im**		**-tineezil-coj**	**-tineezil-am**		'be raspy'
-tapox	**-tapox-im**	**-taptax**	**-taptaj-c**	**-tapo-tam**	**-taptax-lca**	'pull out'

-tim **-c** **-ca** **-lca** **-coj** **-tam** **-cam**

INCREMENTAL MAPPING

Incrementality of the exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):

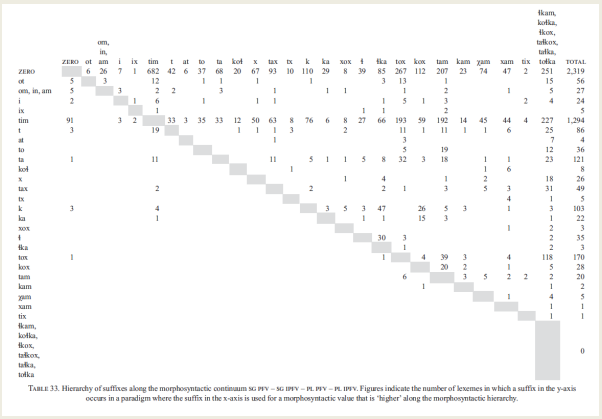


TABLE 33. Hierarchy of suffixes along the morphosyntactic continuum SG PFV – SG IPFV – PL PFV – PL IPFV. Figures indicate the number of lexemes in which a suffix in the y-axis occurs in a paradigm where the suffix in the x-axis is used for a morphosyntactic value that is 'higher' along the morphosyntactic hierarchy.

Incrementality of the features

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):

Features

subject number \ event number	neutral	multiple	distributional
singular	SG NEUTRAL	SG MULT	SG DIST
plural	PL NEUTRQL	PL MULT	PL DIST

singular → plural

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST

Incrementality of the exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):

Exponents

-(t)am,	-ca,	-coj,	-cam,	-c,	-lca,	-(t)im,	...

-(t)im	**-c**	**-ca**	**-lca**	**-coj**	**-(t)am**	**-cam**

singular → plural

Incrementality of the features/exponents

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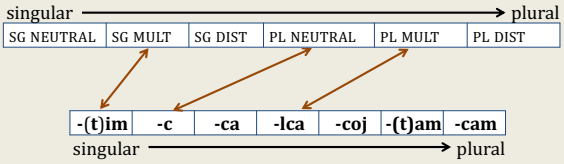
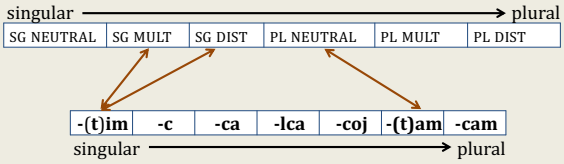


SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-atox	**-atox-im**	**-atyax**	**-atyaj-c**	**-atyax-lca**		'stretch'

31

Incrementality of the features/exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):

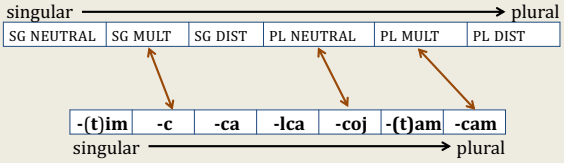


SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-tzam	**-tzam-tim**	**-tzamlqu-im**	**-tzamlc-am**			'corner, attack'

32

Incrementality of the features/exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):

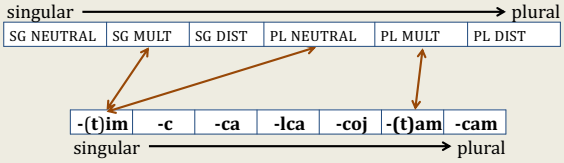


SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-tanamj	**-tanaml-c**		**-tanaml-coj**	**-tanaml-cam**		'hurry'

33

Incrementality of the features/exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):



SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-tpoc	**-tpoct-im**		**-tpocl-im**	**-tpocal-am**		'fall'

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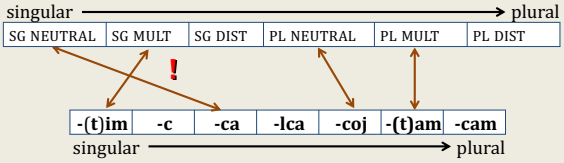
Incrementality of the features/exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):



SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-tineezil-ca	**-tineezil-im**		**-tineezil-coj**	**-tineezil-am**		'be raspy'

-(t)im is an outlier

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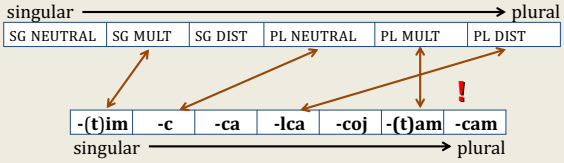
Incrementality of the features/exponents

- The system is not completely random however, because both the paradigm cells and the exponents *mostly* follow an implicational scale (Baerman 2016):



SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT	PL DIST	
-tapox	**-tapox-im**	**-taptax**	**-taptaj-c**	**-tapot-am**	**-taptax-lca**	'remove'

-(t)am is an outlier

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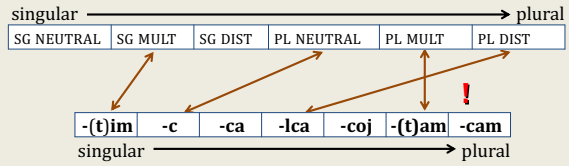
Incrementality and *-tim/-tam*

- Two notable exceptions to this not completely random system are the endings *–tim* and *–tam*



- These are outliers in several respects
  - They are only found with verbs (so far as we know)
  - **-tim/-tam** are among the 5 most frequent exponents
  - **-tim** is most often found marking singular subject MULT forms
  - **-tam** is perceived by speakers as being 'the plural of **-tim**' in morphological elicitation tasks
- Hypothesis: productive verbal suffixes **-tim/-tam** have been overlaid on an older system of number marking shared between verbs and nouns

WIESE 2013

- Wiese 2013 models Latin case morphology and its syncretisms using a incremental scale
  - Case exponents are ordered from least marked to most marked
  - Case values are ordered from least marked to most marked

Table 6 Correlation of formal and functional sequences (*u*-, *ē*-, *a*-, *o*-declensions)

formal sequence (types/markers/endings)										
Types	0	1	2	3	4	5	6	6+	7	7+
Markers	**–**	**s**	**m**	**L**	**vL**	**Ls**	**vLs**	**-X-s**	**vm**	**-X-m**
u-dec.	*-u*	*-us*	*-um*	*-ū*	*-uī*	*-ūs*		*-ibus*	*-uum*	
ē-dec.	——	*-ēs*	*-em*	*-ē*	*-ei*	*-ēs*		*-ēbus*		*-ērum*
a-dec.	*-a*	——	*-am*	*-ā*	*-ae*	*-ās*	*-īs*			*-ārum*
o-dec.	*-e*	*-us*	*-um*	*-ō*	*-ī*	*-ōs*	*-īs*			*-ōrum*
	↕	↕	↕	↕	↕	↕	↕		↕	
u-dec.	VNA_{Ntr}	VN	A	Ab	D	G VNA_{Pl}	AbD_{Pl}		G_{Pl}	
ē-dec.	——	VN	A	Ab	DG	VNA_{Pl}	AbD_{Pl}		G_{Pl}	
a-dec.	VN	——	A	Ab	DG VN_{Pl}	A_{Pl}	AbD_{Pl}		G_{Pl}	
o-dec.	V	N	VNA_{Ntr}/A	AbD	G VN_{Pl}	A_{Pl}	AbD_{Pl}		G_{Pl}	
functional sequences (syncretism fields)										

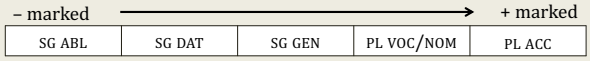
VNA_{NtrPl} not included

- Wiese 2013 models Latin case morphology and its syncretisms using a incremental scale
  - Case exponents are ordered from least marked to most marked
  - Case values are ordered from least marked to most marked

– marked → + marked

	V	N	A	Ab	D	G	V_{pl} N_{pl}	A_{pl}	Ab_{pl} D_{pl}	G_{pl}
u-decl. NTR	0			3	4	5			6	7
u-decl.	1		2	3	4	5			6	7
ē-decl	1		2	3	4		5		6	7
a-decl.	0		2	3	4			5	6	7
o-decl. NTR	2			3	4			5	6	7
o-decl.	0	1	2	3	4			5	6	7

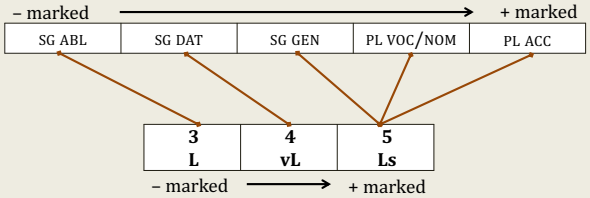
- Wiese 2013 models Latin case morphology and its syncretisms using a incremental scale
  - Case exponents are ordered from least marked to most marked
  - Case values are ordered from least marked to most marked
- We present Weise 2013 using the notation we have been using for Seri



3 L	4 vL	5 Ls

– marked → + marked

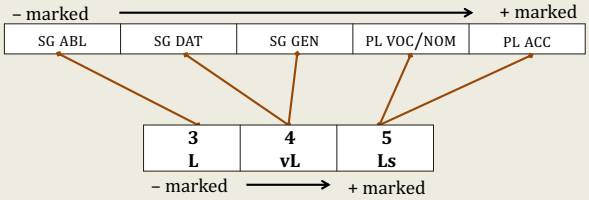
Latin per Wiese (2013)



u-declension

SG ABL	SG DAT	SG GEN	PL VOC/NOM	PL ACC	
ictu:	**ictui:**	**ictu:s**	**ictu:s**	**ictu:s**	'a blow'

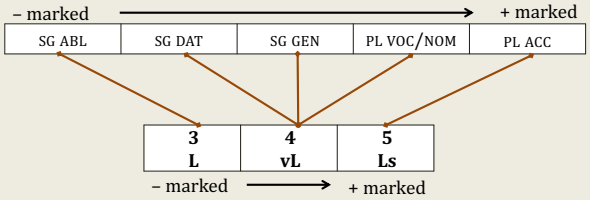
Latin per Wiese (2013)



ē-declension

SG ABL	SG DAT	SG GEN	PL VOC/NOM	PL ACC	
die:	**diei**	**diei**	**die:s**	**die:s**	'a day'

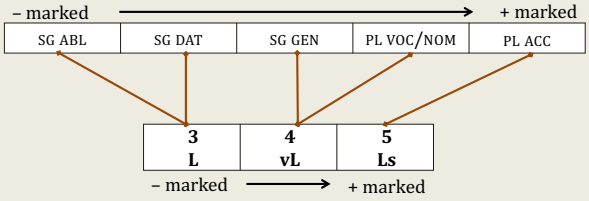
Latin per Wiese (2013)



a-declension

SG ABL	SG DAT	SG GEN	PL VOC/NOM	PL ACC	
capra:	**caprae**	**caprae**	**caprae**	**capra:s**	'a goat'

Latin per Wiese (2013)



o-declension

SG ABL	SG DAT	SG GEN	PL VOC/NOM	PL ACC	
lupo:	**lupo:**	**lupi:**	**lupi:**	**lupo:s**	'a wolf'

CONCLUSION

- Both Wiese (2013) and we have argued for the existence of morphological hierarchies as a paradigmatic organizing principle, in parallel to a hierarchical arrangement of features.
- But can this be formally implemented? Maybe it's wrong to try and implement this in terms of 'constructive' rules.
- Rather – for Seri at least – the hierarchy represents the diachronic residue of various layers of concatenative plural formation, anchored to a greater or lesser degree in the lexicon, and adapted by analogy to expand the paradigm when needed.
- Two arguments in favor:

- Argument #1: The system shows the diachronic traces of having been build up through the concatenation of plural suffixes, though there's no good reason to think it remains synchronically active.

SG NEUTRAL	PL NEUTRAL	
tooxi	**tooxya-tolca**	'die (animal)'

itap	**itap-t**	'dig (using instrument)'
itap-t	**itap-to**	'pull out (soft or loose things)'
itataaho	**itataaho-j**	'prepare'
intitoozi	**intitoozi-toj**	'release'
itcacj	**itcaczi-l**	'chop'
itamaax	**itamaaj-c**	'make alcoholic beverage'
tiimj	**tiiml-ca**	'clear'
cöitsitox	**cöitsitox-lca**	'strike (match)'

- Argument #2: The rest of the (quite large) verbal paradigm is build up concatenatively through prefixation. Typologically at least two systems are operating simultaneously.

	'open'
subject nominalization, active	**c-aasax**
subject nominalization, passive	**hap-aasax**
subject nominalization, negative	**im-aasax**
action/instrument nominalization (possessed)	**ih-aasax**
complement nominalization (possessed)	**y-aasax**
infinitive	**ih-aasax**
realis-*t*	**it-aasax**
realis-*t*, passive	**tp-aasax**
realis-*t*, negative	**itcm-aasax**
realis-*yo*	**iy-aasax**
realis-*yo*, negative	**iyom-aasax**
emphatic realis	**ix-aasax**
emphatic realis, negative	**ixom-aasax**
realis-*mi* tense	**im-aasax**
independent irrealis	**is-aasax**
independent irrealis negative	**iscm-aasax**
imperative	**h-aasax**
imperative negative	**cm-aasax**

etc. etc.

Table from Moser & Marlett (2010)

Does such a system allow for more flexibility in creating/accommodating new forms/meanings?

- Recent fieldwork has made clear that Seri plural marking allows a relatively high amount of flexibility in the creation/interpretation of verb forms
- Speakers were given 3 judgments in morphological elicitation:
  - "it's definitely a word":
  - "it's definitely not a word": *
  - "maybe it's a word: I understand its meaning but I'm not sure I or others use it" : ?
- Many (but not all) invented words received a ? or ✓ judgement with a plausible proposed meaning ([CON] stands for constructed by linguist, [ELAB] for proposed examples by speakers)

Does such a system allow for more flexibility in creating/accommodating new forms/meanings?

Geno quih yaaco.
Geno DET [3]RLYO.have_house
'Geno has a house.' [CON] [EDSEI7NOV2019DRPM.LKPH.ATHF.GHF.GH]

Geno quih yaacotim.
Geno DET [3]RLYO.have_house.MULT
'Geno is fixing up her house.' [CON] [EDSEI7NOV2019DRPM.LKPH.ATHF.GHF.GH]

*Geno quih ?yaacötj. / *yaacöt.*
Geno DET [3]RLYO.have_house.DIST
'Geno has many houses.' [CON] [EDSEI12NOV2019DRPM.GH.ATHF.LKPH.AMMO]
SC on yaacötj: maybe, if she has many houses

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT / DIST?	
-aaco [CON]	**-aaco-tim [CON]**	***-aacöt [CON]** **?-aacötj [CON]**	**-aacöt [ELAB]** **-aacötj [CON]**	**-aacötolca [ELAB]** **?-aacotam [ELAB]** **-aacojam [ELAB]**	'have (build) a house'

Does such a system allow for more flexibility in creating/accommodating new forms/meanings?

Comcaac coi yaacöt.
Geno DET.PL [3]RLYO.have_house.PL
'The people have a house/houses.' [ELAB] [EDSEI7NOV2019DRPM.LKPH.ATHF.GHF.GH]

Comcaac coi yaacötj.
Geno DET.PL [3]RLYO.have_house.PL
'The people have a house/houses.' [ELAB] [EDSEI7NOV2019DRPM.LKPH.ATHF.GHF.GH]

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT / DIST?	
-aaco [CON]	**-aaco-tim [CON]**	***-aacöt [CON]** **?-aacötj [CON]**	**-aacöt [ELAB]** **-aacötj [CON]**	**-aacötolca [ELAB]** **?-aacotam [ELAB]** **-aacojam [ELAB]**	'have (build) a house'

Does such a system allow for more flexibility in creating/accommodating new forms/meanings?

Comcaac coi yaacötolca.
Geno DET.PL [3]RLYO.have_house.PL.MULT
'The people have fixed up a house/houses.' [ELAB] [EDSEI7NOV2019DRPM.LKPH.ATHF.GHF.GH]

? *Comcaac coi yaacotam.*
Geno DET.PL [3]RLYO.have_house.PL.MULT
'The people have fixed up a house/houses.' [ELAB] [EDSEI7NOV2019DRPM.LKPH.ATHF.GHF.GH]
SC: it's a new word, if it's many people who fixed up their house.

Comcaac coi yaacojam.
Geno DET.PL [3]RLYO.have_house.PL.MULT
'The people have been building houses.' [ELAB] [EDSEI12NOV2019DRPM.GH.ATHF.LKPH.AMMO]
SC: if it's people building many houses: they finish one then build another

SG NEUTRAL	SG MULT	SG DIST	PL NEUTRAL	PL MULT / DIST?	
-aaco [CON]	**-aaco-tim [CON]**	***-aacöt [CON]** **?-aacötj [CON]**	**-aacöt [ELAB]** **-aacötj [CON]**	**-aacötolca [ELAB]** **?-aacotam [ELAB]** **-aacojam [ELAB]**	'have (build) a house'

¡Haa xah tiipe!